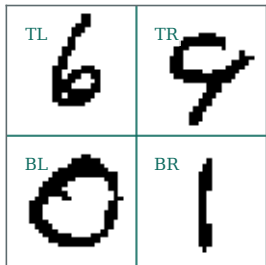
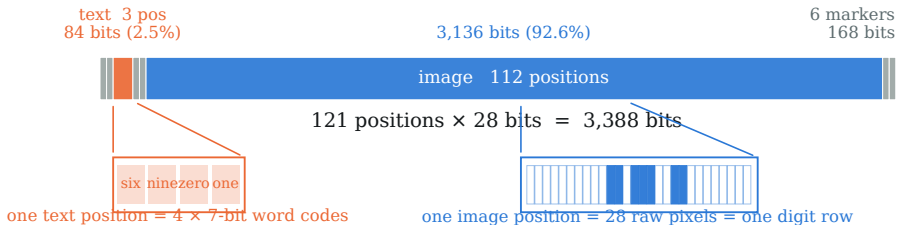


(a) one example

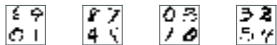


56×56, binary pixels
native 28×28 digits, never resized

(b) written into one bitstream — no tokenizer on either side



(c) image → text



given the image, the model writes:

“six + nine + zero + one = sixteen”

97.90% exactly correct

(d) text → image



drawn from the equation alone:

“six + nine + zero + one = sixteen”

99.98% all four digits right

(e) joint — both from noise



its own equation, same trajectory:

“seven + nine + four + zero = twenty”

96.85% mutually consistent